The installation script should ask if all users on the server will use the same database. During the initial installation, the question is asked. If yes, a single sqlite file should be used. The question asking where to save messages in subsequent installations for clients can prompt the admin with a confirmation of the save location.

The installation for a subsequent client can ask if a single save location should be used, if it isn't already. If the answer is yes, all independent sqlite installations should be merged into a single database at the supplied location. Afterwards, any installations for new clients would prompt the admin with a notification of the save location.

How are the two different ways of running a database different? If all clients use the same database, perhaps the order of messages could be seen. Messages sent to more than one client, and subscriptions to the primary contact point could be stored as one message.

How the Installation Script Works

The updated `install.sh` and the newly added `merge_databases.php` utility implement the exact installation and migration workflow:

```
flowchart TD
    Start["Run sudo ./install.sh"] --> CheckInstalls{"Any prior installations
registered?"}

    CheckInstalls -->|No: Initial Install| PromptInitial["Prompt: Will all users on
the server use the same database? [Y/n]"]
    PromptInitial -->|Yes| SetShared["Prompt for / confirm single save
location<br/>(/var/lib/tricurse/contact/contact.sqlite)<br/>Save server.json
(shared_database: true)"]
    PromptInitial -->|No| SetIndep["Prompt for domain storage
directory<br/>(/var/lib/tricurse/contact/domain/)<br/>Save server.json
(shared_database: false)"]

    CheckInstalls -->|Yes: Subsequent Install| CheckShared{"Is server already using
a single database?"}
    CheckShared -->|Yes| PromptConfirm["Prompt admin with confirmation of the save
location<br/>'Confirm save location [/path/to/contact.sqlite]? [Y/n]'"]
    CheckShared -->|No: Independent DBs Exist| PromptConvert["Prompt: Should a
single save location be used for all clients on this server instead? [y/N]"]

    PromptConvert -->|Yes| RunMerge["1. Prompt for single save location<br/>2. Run
merge_databases.php<br/>3. Merge all independent SQLite DBs<br/>4. Update all
webroot db_config.php files<br/>5. Save server.json (shared_database: true)"]
    RunMerge --> NotifyAfter["Afterwards: All future new client
installations<br/>prompt admin with notification of save location"]
    PromptConvert -->|No| KeepIndep["Keep independent DB for this client"]
```

1. Initial Installation

- The installer checks if any installations are registered on the server.
- If none exist, it asks:

```
Will all users on the server use the same database? [Y/n]:
```

- **If Yes**: Prompts to confirm or customize the single save location (default `/var/lib/tricurse/contact/contact.sqlite`), writes the server configuration to `/etc/tricurse/server.json`, and initializes the schema on the single database.
- **If No**: Configures independent database mode and prompts for the domain's dedicated message storage folder.

2. Subsequent Client Installation (Already Using a Single Database)

- When a single database is already active, the installer prompts the administrator with a confirmation:

```
This server is configured to use a single shared database for all clients.
Configured save location: /var/lib/tricurse/contact/contact.sqlite
Confirm save location [/var/lib/tricurse/contact/contact.sqlite]? [Y/n] (or
specify new path):
```

3. Subsequent Client Installation (Converting Independent to Single Shared)

- If independent databases were previously used, the installer asks:

```
Currently, installations on this server use independent databases.
Should a single save location be used for all clients on this server instead?
[y/N]:
```

- **If Yes**:
    1. Prompts for the unified target location (default `/var/lib/tricurse/contact/contact.sqlite`).
    2. Executes `merge_databases.php`, copying and consolidating all accounts, messages, settings, blocklists, and push subscriptions from each independent database file with zero duplicate message bodies.
    3. Updates each existing client's webroot `db_config.php` and `/etc/tricurse/<domain>/contact.conf`.
    4. Updates `server.json` to mark the server as shared.
    5. Afterwards, all future client installations notify and confirm the admin of the unified save location.

How the Two Database Modes Differ

Feature	Single Shared Database	Independent Databases
Storage & Deduplication	**Single copy stored on disk**: Multi-recipient messages and broadcasts from the primary contact point write a single row in `incoming_messages`.	**Duplicated copies**: Every recipient domain stores its own copy in its own `.sqlite` file.
Global Message Ordering	**Unified chronological timeline**: All messages across all domains share a single auto-incrementing ID sequence and unified `received_at` / `sent_at` timeline.	**Fragmented order**: Each database has its own ID sequence starting from 1. Cross-domain sequence can only be reconstructed by parsing multiple files.
Primary Site Subscriptions	**Zero HTTP overhead & zero disk bloat**: Local clients subscribed to the main contact point read the broadcast record directly using per-domain read/delete tracking (`message_reads`, `message_deletions`).	**Multiple writes**: The main point must loop and dispatch or duplicate records into each client's separate file.
Backups & Maintenance	**One-step operations**: A single backup of `contact.sqlite` covers all domains on the server; migrations and updates run in a single pass.	**Multi-step operations**: Backing up or migrating requires touching each domain's separate SQLite directory.
Tenant Isolation & Security	Data is partitioned logically by `domain` and `target_domain` in SQL queries. All web processes share read/write access to the database directory.	Strong filesystem boundary: Each `.sqlite` file can be isolated under separate POSIX users or isolated PHP-FPM pools.
Portability	Exporting a single client requires filtering rows by domain.	Exporting a client simply requires copying their individual `contact.sqlite` file.

Updated Files

- `install.sh`: Guided initial question, subsequent confirmation, and migration prompts.
- `merge_databases.php`: Automated multi-database consolidation utility (also installed as CLI utility `merge_databases`).

- `contact_list.php`: Displays whether each registered site uses an independent or `Shared` database file.
- Synchronized to deployment locations in `/addr.bar/addressbar.net/contact/` and `/var/www/html/contact/`.